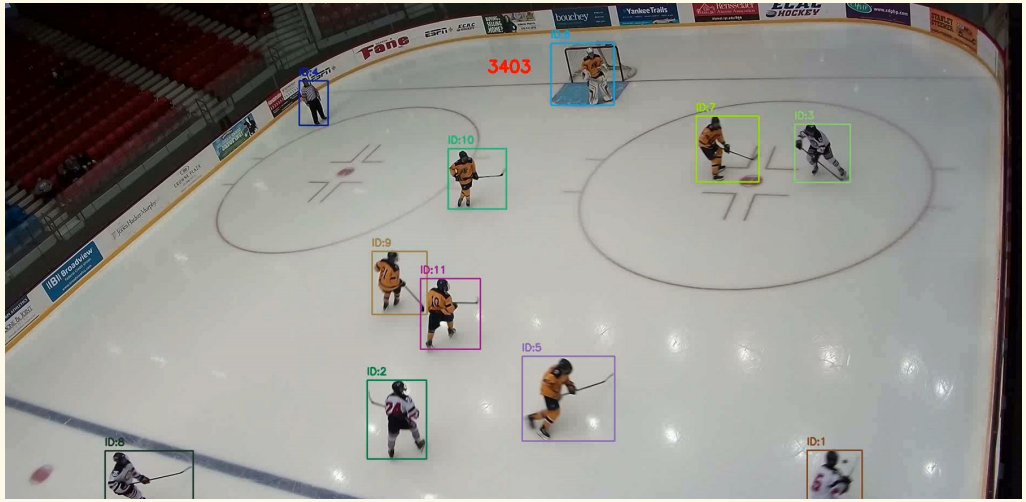
Evaluating Multi-Object Trackers for Collegiate Hockey



Henry Robb

Agenda

Introduction

Related Work

Methodology

Experimental Setup

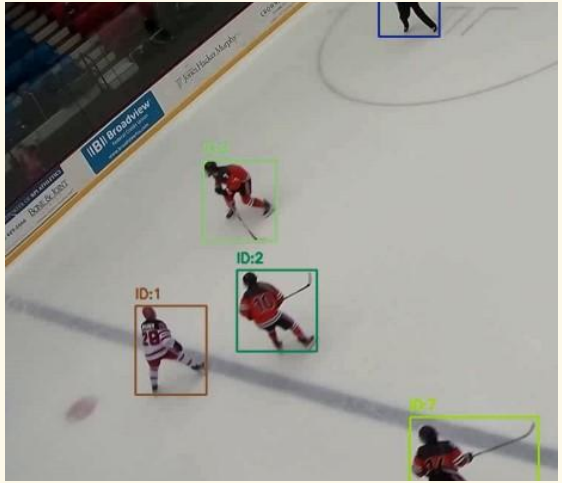
Results

Discussion and Next Steps

Conclusion

Introduction

V1	V2	V3	V4	V5	H1	H2	H3
(238, 800)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(628, 974)	(394, 754)	(339,
(239, 794)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(634, 967)	(397, 746)	(342,
(240, 788)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(640, 961)	(401, 738)	(345,
(246, 779)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(647, 955)	(403, 731)	(348,
(245, 774)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(658, 945)	(406, 723)	(350,
(243, 769)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(667, 939)	(408, 716)	(353,
(241, 764)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(673, 932)	(409, 709)	(355,
(239, 760)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(678, 925)	(410, 701)	(359,
(236, 755)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(683, 917)	(411, 695)	(362,
(233, 752)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(687, 909)	(412, 688)	(365,
(229, 749)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(689, 903)	(411, 683)	(368,
(222, 747)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(691, 884)	(411, 676)	(372,
(217, 746)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(696, 877)	(410, 671)	(380,
(213, 743)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(703, 876)	(409, 666)	(380,
(208, 741)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(708, 869)	(408, 661)	(388,
(204, 739)	(-1, -1)	(-1, -1)	(-1, -1)	(-1, -1)	(713, 862)	(405, 655)	(393,
(201, 735)	(98, 658)	(-1, -1)	(-1, -1)	(-1, -1)	(716, 854)	(403, 650)	(398,
(194, 735)	(108, 650)	(-1, -1)	(-1, -1)	(-1, -1)	(719, 846)	(400, 646)	(403,
(190, 732)	(118, 642)	(-1, -1)	(-1, -1)	(-1, -1)	(722, 836)	(398, 642)	(408,
(186, 730)	(124, 630)	(-1, -1)	(-1, -1)	(-1, -1)	(726, 828)	(395, 638)	(413,
(172, 716)	(135, 623)	(-1, -1)	(-1, -1)	(-1, -1)	(729, 820)	(392, 635)	(419,
(169, 713)	(146, 615)	(-1, -1)	(-1, -1)	(-1, -1)	(733, 813)	(389, 631)	(424,
(175, 707)	(161, 617)	(-1, -1)	(-1, -1)	(-1, -1)	(737, 805)	(385, 628)	(429,
(161, 699)	(173, 603)	(-1, -1)	(-1, -1)	(-1, -1)	(741, 799)	(382, 625)	(427,
(162, 699)	(183, 594)	(-1, -1)	(-1, -1)	(-1, -1)	(745, 792)	(378, 622)	(431,
(161, 695)	(194, 585)	(-1, -1)	(-1, -1)	(-1, -1)	(750, 786)	(374, 619)	(434,
(159, 692)	(204, 582)	(-1, -1)	(-1, -1)	(-1, -1)	(754, 778)	(370, 616)	(437,
(157, 685)	(213, 567)	(-1, -1)	(-1, -1)	(-1, -1)	(759, 771)	(366, 613)	(440,
(157, 681)	(224, 566)	(-1, -1)	(-1, -1)	(-1, -1)	(764, 764)	(363, 611)	(443,
(157, 677)	(225, 549)	(-1, -1)	(-1, -1)	(-1, -1)	(769, 757)	(359, 609)	(447,
(158, 672)	(232, 538)	(-1, -1)	(-1, -1)	(-1, -1)	(774, 750)	(355, 606)	(450,



Professional Software

High budget and use of infrared chips

Manual Annotation

Low budget but very inefficient

Automatic Tracking

Low budget and effeicient

Related Work

Occlusion

Speed Change

Motion Blur

Related Work: Trackers

ByteTrack

Bridges occlusion gaps based off pure motion

High processing speed baseline

StrongSORT

Uses deep learning visual extractors to identify players

HybridSORT

Uses weak cues (YOLO confidence scores) to maintain identity

BoTSORT

Directly estimates bounding box width and height (instead of aspect ratio)

Can compensate for camera motion

Methodology

Datasets

HockeyAI: ~2100 annotated hockey photos

MSMT17: Pedestrian dataset on which the Re-ID backbones were trained

RPI Hockey Footage: Game footage of the NCAA Division I Men's and Women's hockey teams

Libraries

Ultralytics: provided YOLOv8 for object detection

Boxmot: provided ByteTracker, StrongSORT, HybridSORT, and BoTSORT for tracking

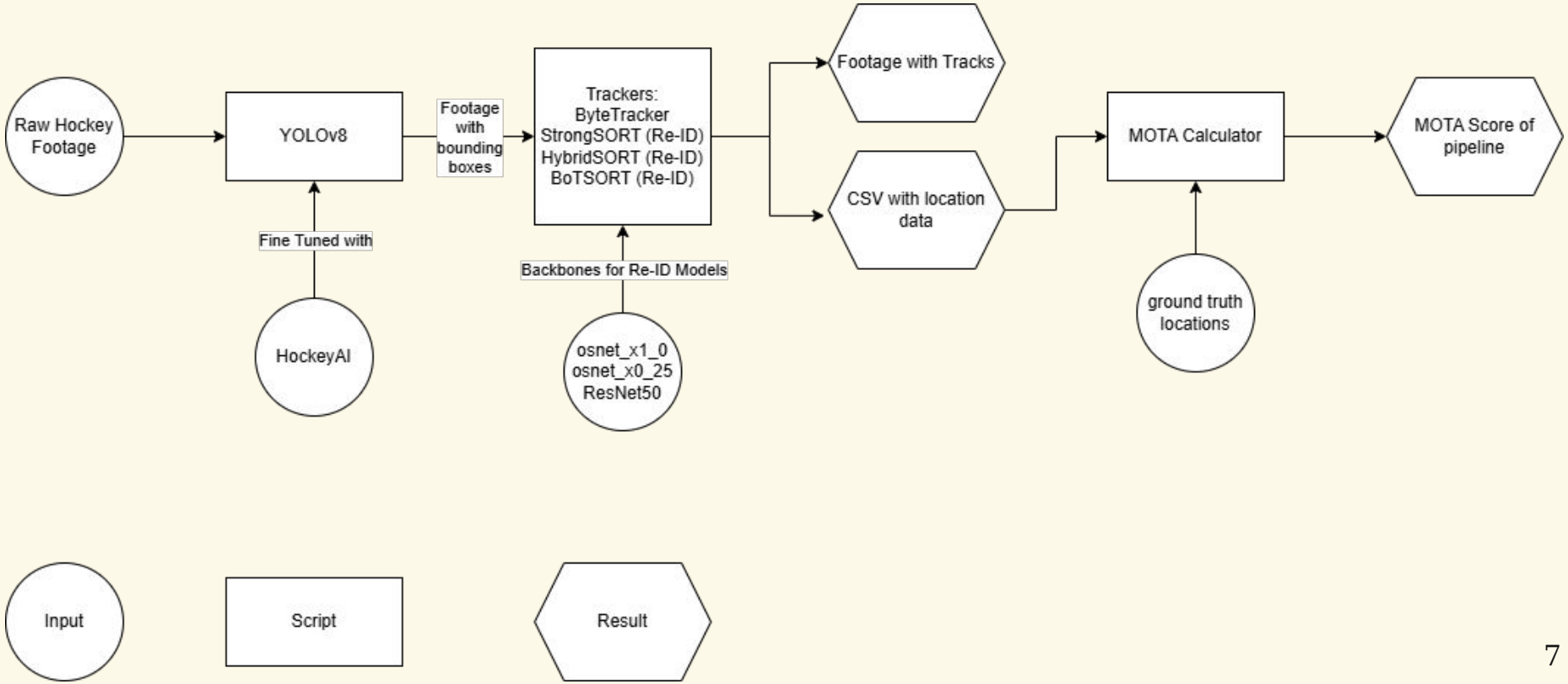
Evaluation

FPS was calculated to determine efficiency

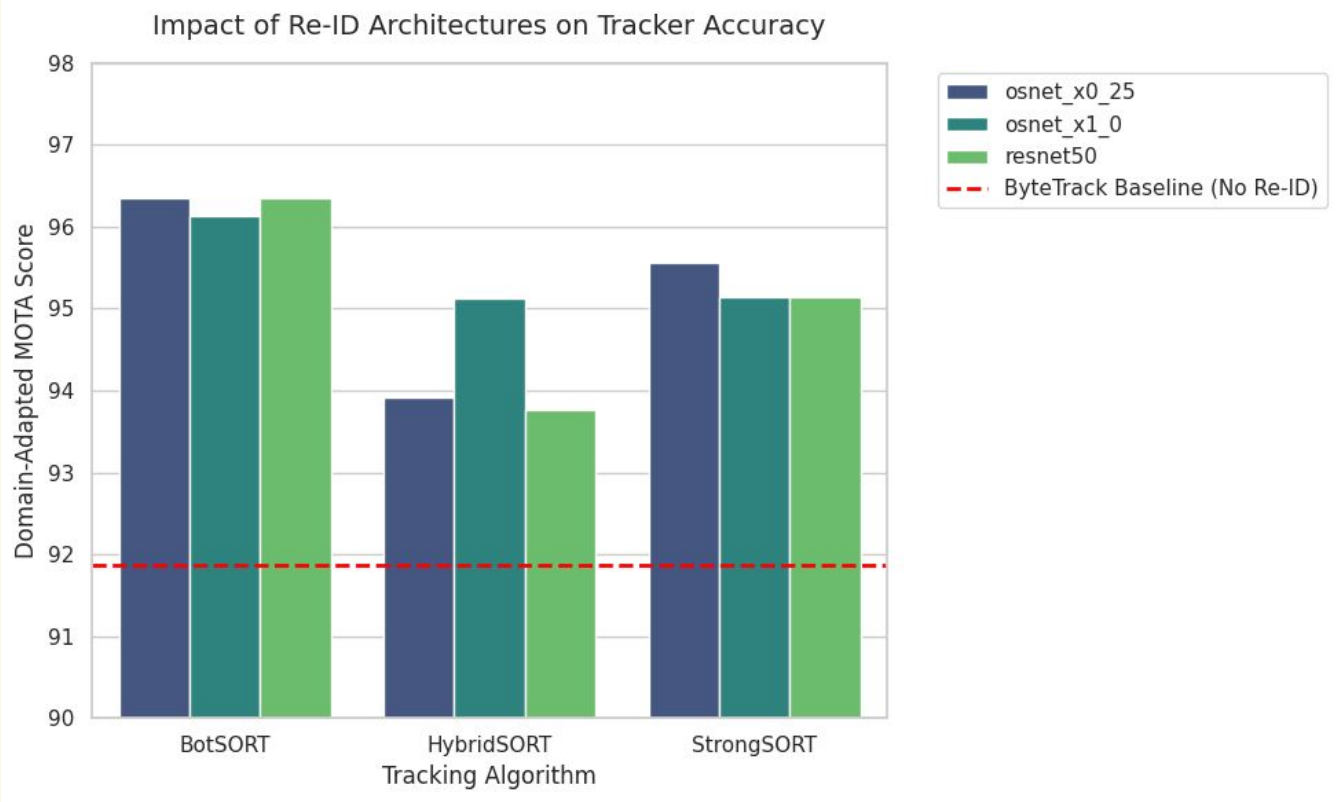
Used a domain adapted Multi-Object Tracking Accuracy (MOTA) on the outputs

$$MOTA = 1 - \frac{FN + IDSW}{GT_{total}}$$

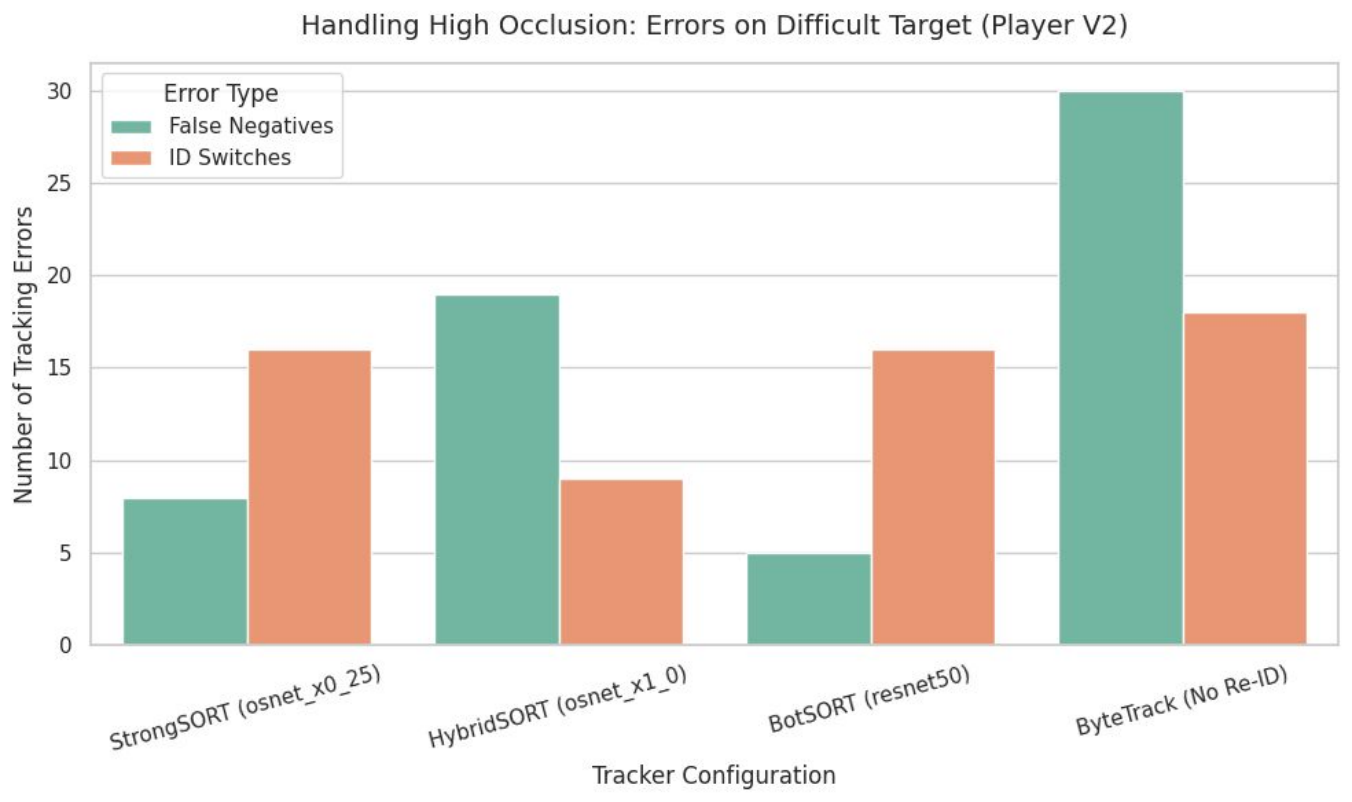
Experimental Setup



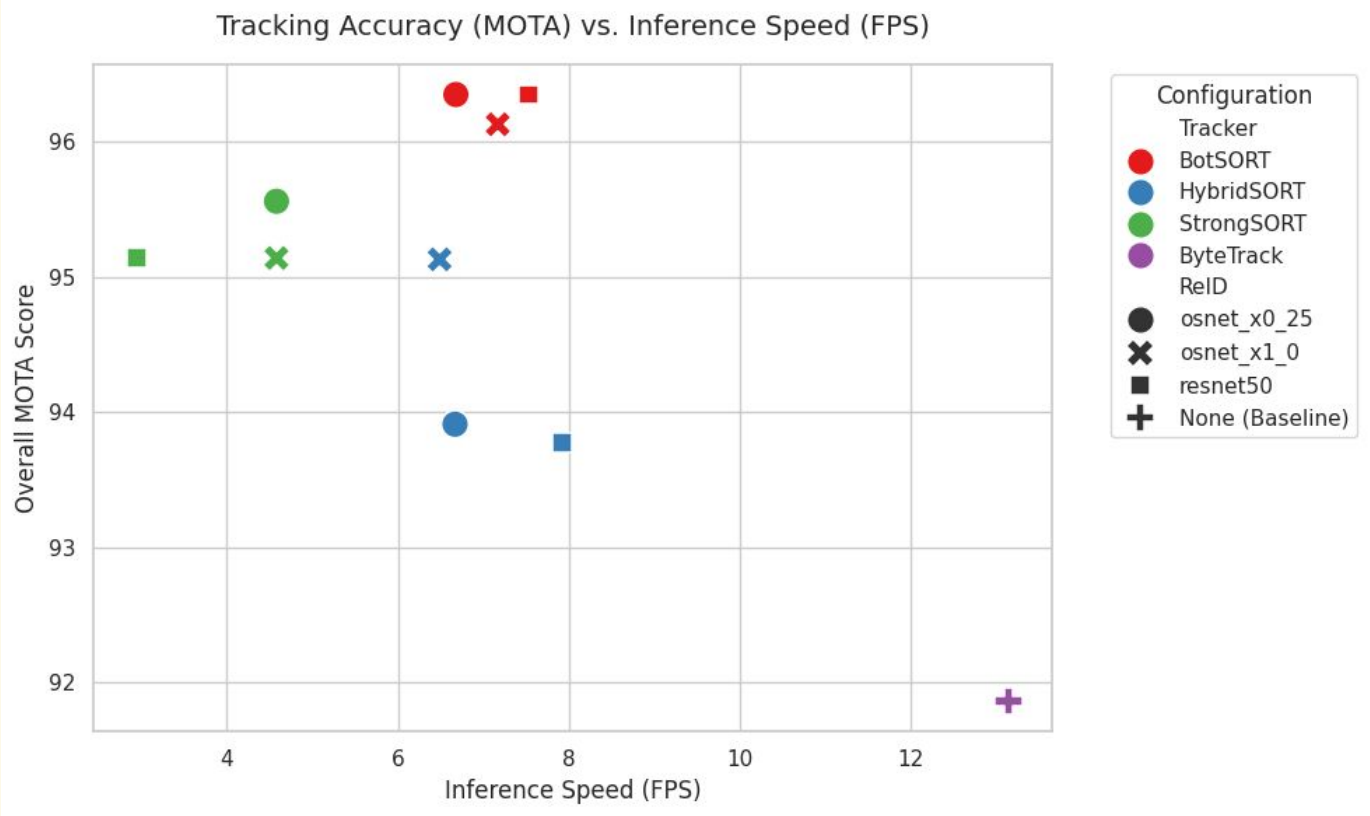
Results


Impact of Re-ID Architectures on Tracker Accuracy
Domain-Adapted MOTA Score
98
97
96
95
94
93
92
91
90
BotSORT
HybridSORT
StrongSORT
Tracking Algorithm
osnet_x0_25
osnet_x1_0
resnet50
ByteTrack Baseline (No Re-ID)

Results


Handling High Occlusion: Errors on Difficult Target (Player V2)
Error Type
False Negatives
ID Switches
Number of Tracking Errors
0
5
10
15
20
25
30
StrongSORT (osnet_x0_25)
HybridSORT (osnet_x1_0)
BotSORT (resnet50)
ByteTrack (No Re-ID)
Tracker Configuration

Results


Tracking Accuracy (MOTA) vs. Inference Speed (FPS)
Overall MOTA Score
96
95
94
93
92
4
6
8
10
12
Inference Speed (FPS)
Configuration
Tracker
BotSORT
HybridSORT
StrongSORT
ByteTrack
ReID
osnet_x0_25
osnet_x1_0
resnet50
None (Baseline)

Discussion

Spatial Architecture's Success

BoT-SORT's success shows that directly estimating width/height is the best defense against aspect ratio warping

Visual Uniformity Affect

Despite general success, no pipeline was able to overcome the challenge of identical jerseys

Hardware Speed Anomaly

The heavier ResNet50 outperformed the lighter OSNet models (BoT-SORT and Hybrid-SORT)

Optimal Pipeline

BoT-SORT paired with ResNet50
MOTA: 96.35
FPS: 7.53

Future Work

Custom Re-ID

Train Re-ID models on hockey players, specifically jersey numbers

Data Scarcity

Ground truth data is painstaking to create, so it will take time to build up a suitable supply

Hardware Upgrades

Using cameras with higher resolution, or with the ability to pan to reduce distortion

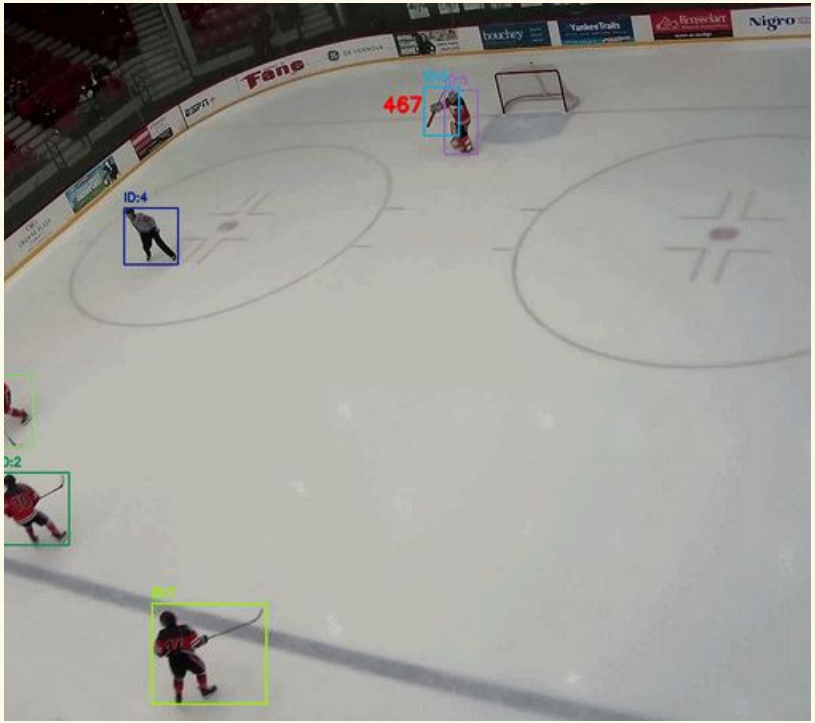
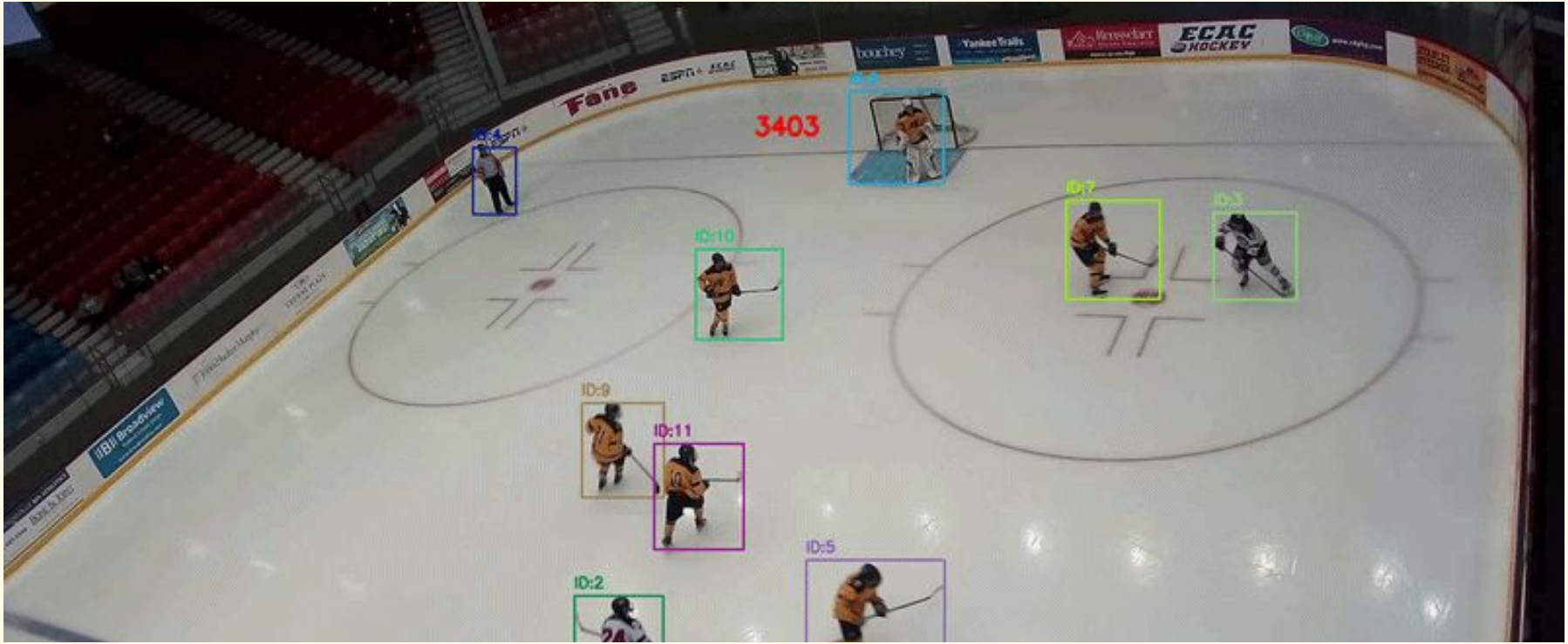
Conclusion

Successfully built an automated pipeline for tracking players

The trackers, despite being built on pedestrian datasets, were effective

The trackers are not quite real time processors

Because of uniform jerseys and occlusions, there are limits on the success



Thank you